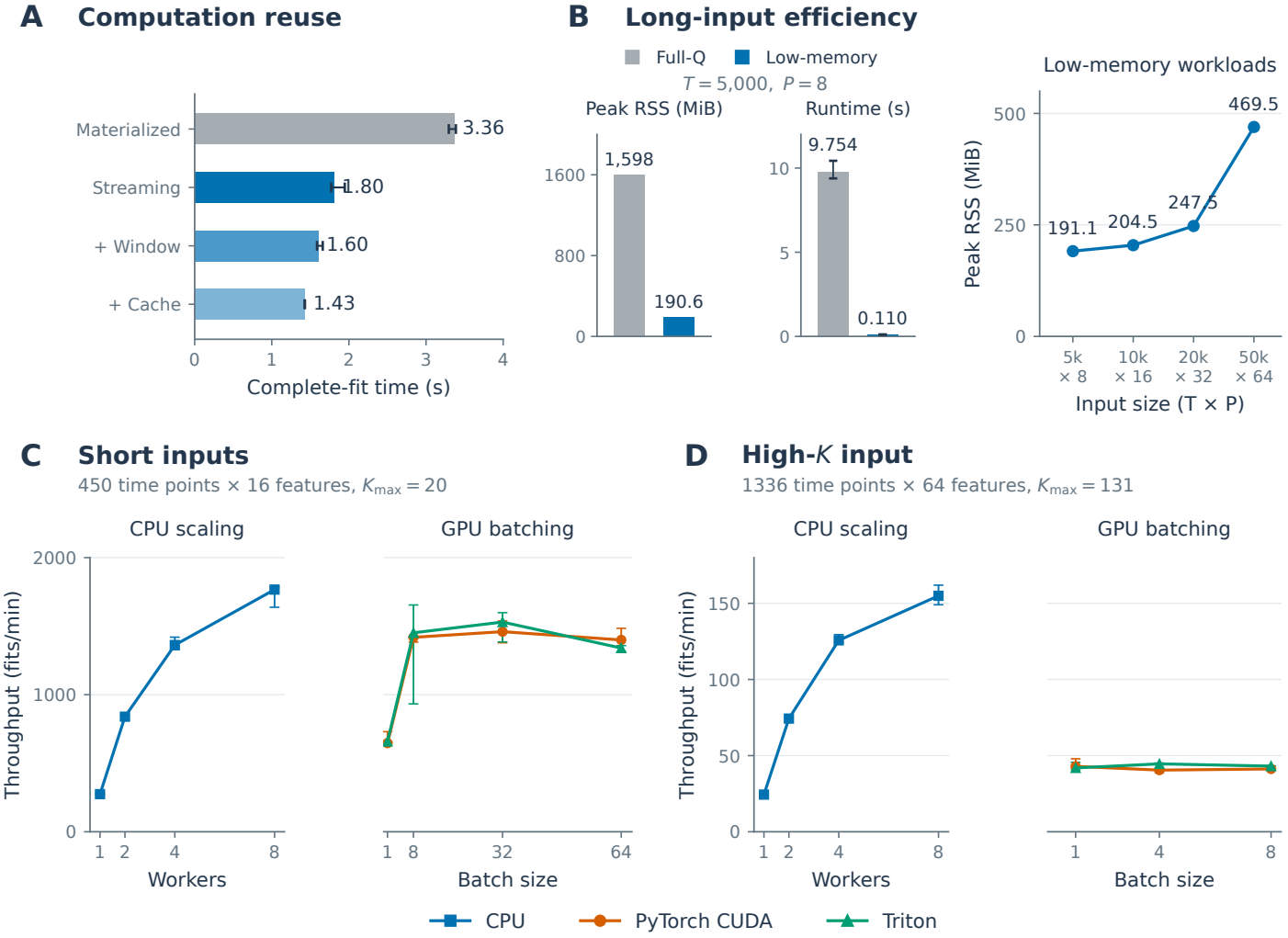
A
Computation reuse
Materialized
Streaming
+ Window
+ Cache
3.36
1.80
1.60
1.43
0
1
2
3
4
Complete-fit time (s)
B
Long-input efficiency
Full-Q
Low-memory
T = 5,000, P = 8
Peak RSS (MiB)
Runtime (s)
1,598
190.6
9.754
0.110
1600
800
0
10
5
0
Low-memory workloads
Peak RSS (MiB)
500
250
0
191.1
204.5
247.5
469.5
5k × 8
10k × 16
20k × 32
50k × 64
Input size (T × P)
C
Short inputs
450 time points × 16 features, K_max = 20
CPU scaling
GPU batching
Throughput (fits/min)
2000
1000
0
1
2
4
8
Workers
1
8
32
64
Batch size
D
High-K input
1336 time points × 64 features, K_max = 131
CPU scaling
GPU batching
Throughput (fits/min)
150
100
50
0
1
2
4
8
Workers
1
4
8
Batch size
CPU
PyTorch CUDA
Triton